Marine Pollution Bulletin 77 (2013) 412–417

Contents lists available at ScienceDirect

Marine Pollution Bulletin

journal homepage: www.elsevier.com/locate/marpolbul

Baseline

Legacy contaminant bioaccumulation in rock crabs in Sydney Harbour during remediation of the Sydney Tar Ponds, Nova Scotia, Canada

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ARTICLE INFO

Keywords:
Rock crabs (*Cancer irroratus*)
Chemical contaminants
Remediation
Sydney Tar Ponds

ABSTRACT

Concentrations of PAHs, PCBs, metals and lipids in hepatopancreas of rock crabs (*Cancer irroratus*) were measured in Sydney Harbour (SH) for one year prior to remediation and three years of remediation of the Sydney Tar Ponds (STP), Nova Scotia. Low level concentrations of PCBs and metals were measured, although PAHs were mostly undetected. Metal concentrations showed little spatio-temporal variability, although highest concentrations of As, Cd and Cu were measured at reference stations furthest from the STP remediation site. Mercury concentrations were at least an order of magnitude lower than Canadian guidelines. Moderately elevated PCB concentrations were detected in crabs near Muggah Creek, but these were generally not higher than those measured during baseline. Despite remediation activities, current contaminant burdens measured in crabs were much lower than previously reported in other studies of crabs and lobster in industrial harbours in eastern Canada, due in part to natural recovery of SH sediments.

Sydney Harbour (SH), Nova Scotia has been subject to effluent and atmospheric inputs of contaminants, including metals, polycyclic aromatic hydrocarbons (PAHs) and polychlorinated biphenyls (PCBs), from a large coking and steel manufacturing facility which operated in Sydney from 1901 until closure in 1988 (Lambert et al., 2006). Contaminant were discharged into the Sydney Tar Ponds (STP) forming a tidal tributary, Muggah Creek. Thousands of tons of metals, PCB and PAH contaminated sludge were released into Muggah Creek and SH (Furinsky, 2002). Many contaminants were particle reactive binding to fine grained, organic-rich harbour sediments with negative ecological health impacts for benthic organisms (Stewart et al., 2002; Tay et al., 2003). Contaminant concentrations in sediments were highest near Muggah Creek, decreasing towards the outer harbour (Lee et al., 2002; Walker et al., 2013a,b). As a result elevated contaminants in SH were reported in American lobster (*Homarus americanus*) (Uthe and Musial, 1986; King et al., 1993; Ernst et al., 1999) leading to closure of the commercial lobster fishery in SH in 1982 (Hilderbrand, 1982; King et al., 1993).

Crabs have been used successfully as biological indicators of marine pollution because they have limited ability to metabolize contaminants found in surface sediments (Firat et al., 2008). Sediments reflect temporal contaminant inputs that crabs can bioaccumulate via prey consumption, depending on sediment burial, bioturbation rates, age of crabs, feeding rates, food preferences, uptake, excretion, and growth rates (Bierman, 1990; Yunker and Cretney, 1996). Crabs are bottom scavengers and their diet consists of organisms of different trophic levels, including clams, mussels (*Mytilus* spp.), polychaetes, smaller crustaceans and fish, but will feed opportunistically on fish carcasses. Contaminants are stored in fat-rich digestive glands (hepatopancreas) and bioaccumulate at higher levels than finfish or other shellfish, partly because crabs are relatively sedentary and because they favour substrates where contaminated sediments accumulate (Chou et al., 2003). Absorption of waterborne contaminants via gills has been observed by Spacie and Hamelink (1985), so becomes another pathway for contaminants to accumulate in crabs. Contaminants in seawater were often below instrument detection limits (DLs) (Walker et al., 2013c), therefore, rock crabs (*Cancer irroratus*) were chosen as suitable ecological indicators to measure changes in contaminant concentrations in SH and because they are ideal surrogates for contaminants accumulating in lobster (Chou et al., 2002). The objectives of this study were: (1) to assess potential impacts of remediation activities in the marine environment by measuring changes of contaminant concentrations in crab hepatopancreas tissue; and (2) to establish a new baseline of chemical contaminants in crab tissues in what was described as Canada's most polluted site.

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0025-326X/$ - see front matter © 2013 Elsevier Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.marpolbul.2013.09.036

The land-based remediation at the STPs began in 2009 using *in situ* solidification and stabilization techniques to immobilize contaminants. Marine stations (~10 m deep), were established and sampled by boat during one baseline year (2009) and three years of remediation (2010, 2011 and 2012) between July 2009 and July 2012 (Fig. 1). Potential seasonal variations were reduced by sampling every year in July when the highest concentrations of chemicals (and seasonally higher lipid contents associated with reproduction) were found in hepatopancreas tissues (Chou et al., 2003). Stations were distributed across four assessment areas to reflect the harbour ecosystem as follows: area 1 – near-field, area 2 – mid-field, area 3 – far-field, and area 4 – Sydney River Estuary and were similarly distributed to a study of sediment contaminants (Walker et al., 2013a,b). Annual Department of Fisheries and Oceans Canada (DFO) annual scientific crab licenses were obtained prior to collection (DFO, 1998).

Baseline surveys revealed indigenous rock crabs were dominant in SH, although some by-catch of non-indigenous European green crabs (*Carcinus maenas*), were often encountered at inner harbour stations. No by-catch of any other crab species, lobster or fish were retained. Only adult male rock crabs were chosen for this study (legal carapace length between 102 and 130 mm). Female crabs were excluded to avoid potential gender biases. Triplicate crab traps, baited with mackerel, were deployed at each station and retrieved 2 d later. Approximately 6–13 crabs from triplicate traps were pooled to obtain minimum weight required for composite hepatopancreas tissue analysis. Catch success were highly variable resulting in a lack of representative samples from each station every year, but at least one composite sample was collected from each assessment area throughout this study. Crabs were transported live on ice in breathable containers within 6 h for dissection and analysis by Maxxam Analytics (accredited through the Standards Council of Canada).

Crab hepatopancreas tissue was analyzed for PAHs (18 individual PAH compounds) based on US-EPA 8270C; aroclor PCBs based on US-EPA US-EPA 8082; metals (As, Cd, Cu, Pb, Zn) based on US-EPA 6020A; Hg analysis was based on US-EPA 245.6; and lipid content based on AOAC 948.16 (US-EPA, 2005). Tissue concentrations were expressed as wet weight. At least one blind field duplicate was included for every ten samples. Significant differences spatially (between areas) and temporally (between baseline and remediation) were determined by one-way ANOVA followed by Tukey's test using Minitab; years attributed with same letters were not significant and those with different letters were significantly different ($p < 0.05$). Baseline data represents "before" and were compared with annual remediation data, representing "after" treatment.

All crabs appeared healthy with no obvious shell deformities. By-catch (lobster and green crabs) were returned unharmed. Pooled hepatopancreas tissue weights ranged from 40 to 160 g. Blind field duplicate results varied from parent sample results ±11% (depending on analyte). Although pooling can reduce effects of variation it is not entirely eliminated due to biological averaging of samples (Correll, 2001; Mary-Huard et al., 2007).

PAH concentrations were not detected (<0.05 $\mu g\ g^{-1}$) during the study, except for a single detection of naphthalene at station 4-1 (0.14 $\mu g\ g^{-1}$) in year 3, despite remediation activities indicating limited impacts of PAHs on biota in the marine environment. For comparison, if PAHs < DLs were assumed to be ½DL (0.025 $\mu g\ g^{-1}$), then PAH concentrations in crab tissue were up

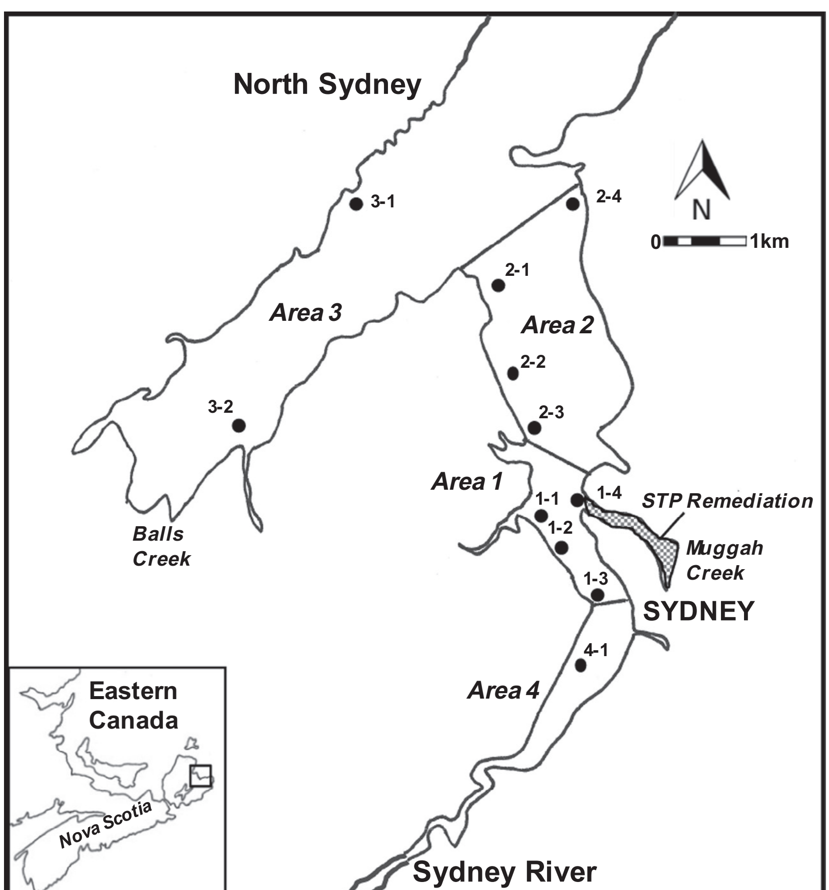

Fig. 1. Marine stations and assessment areas in Sydney Harbour relative to remediation site.

to 520 times lower than those reported by Ernst et al. (1999) who measured PAH concentrations of 13 μg g^{-1} in lobster. Prouse (1994) ranked other harbours in the Maritimes (Nova Scotia, New Brunswick and Prince Edward Island) based on PAH concentrations in lobster and concluded that SH had the highest potential for PAH contamination.

Wild and Tasto (1983) found crab muscle tissues accumulated 3–10 times less PAH load compared to hepatopancreas tissue even in crabs from areas receiving considerable petroleum inputs. They found both adult and large juvenile Dungeness crabs in the San Francisco area carried larger burdens of PAHs in their hepatopancreas tissue with little or no contamination of muscle tissue. Assuming PAHs were not detectable in hepatopancreas tissue, where contaminants are known to bioaccumulate (Chou et al., 2003), the potential for high concentrations of PAHs to accumulate in edible crab muscle tissue may also be low. These results are particularly encouraging for the rock crab fishery in eastern Canada (Gendron et al., 2001; Schaefer et al., 2004), and the American lobster fishery which is also highly dependent on rock crabs as a food source.

Significant spatial variation (histograms) in PCB concentrations were observed between areas 1 and 2 ($F = 6.40$, $p = 0.022$) and between areas 1 and 3 ($F = 24.14$, $p = 0.000$), with area 3 stations

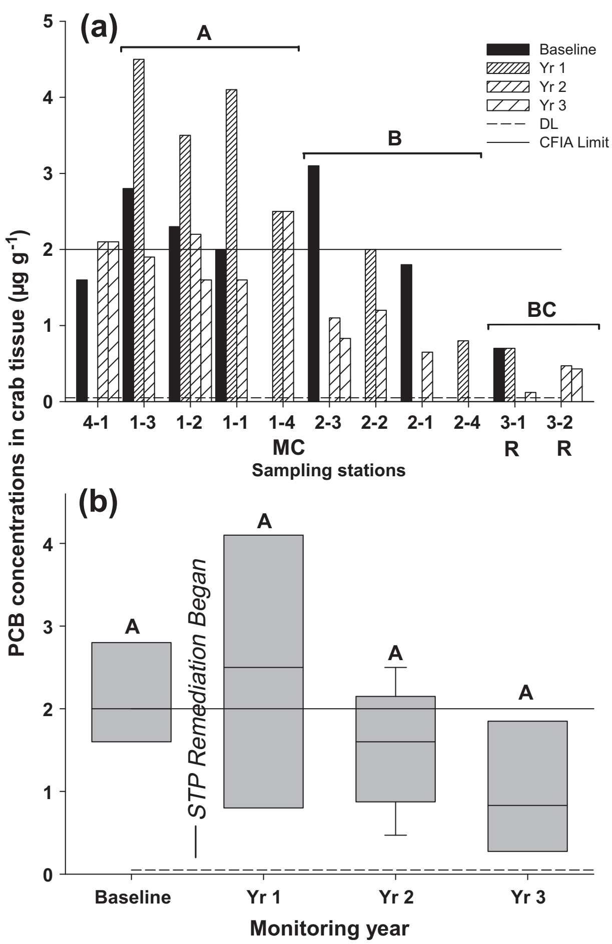

Fig. 2. (a) Concentrations of PCB in crabs during baseline and 3 years of remediation. Dashed line indicates detection limit (DL) (0.05 μg g^{-1}). Canadian Food Inspection Agency (CFIA) limit indicated with solid horizontal line (2 μg g^{-1}). Stations centered on Muggah Creek (MC) and reference stations are indicated (R). (b) PCB concentrations showing temporal variation during baseline and remediation. Significant differences determined using one-way ANOVA followed by Tukey's test; areas or years attributed with same letters were not significant and those with different letters were significantly different (p<0.05).

significantly lower than area 2 ($F = 5.31$, $p = 0.047$) (Fig. 2a). Temporal trends (box plots) show highest annual mean PCB concentrations during year 1, whereas lowest annual mean was during year 3 (Fig. 2b). However, one way ANOVA test results indicate that there were no significant differences between baseline and remediation (year 1, $F = 0.70$, $p = 0.419$; year 2, $F = 1.88$, $p = 0.192$; year 3, $F = 4.72$, $p = 0.055$). Comparisons of PCB concentrations in crab tissue and surface sediment were poorly correlated across all sampling years ($r^2 = 0.445$, $n = 28$). PCB concentrations were higher than baseline at stations 1–1, 1–2, 1–3 during year 1 and 2 remediation, but then returned to concentrations that were below baseline in year 3 which provides some evidence of spatial and temporal transitory exceedances of the CFIA limit. Although comparable annual PCB data did not exist for every station (due to variation in catch success), these limited data highlight the variability of using biota for monitoring, but they still do indicate that PCB concentrations in crabs during STPs remediation were stable compared to baseline conditions.

The highest As, Cd and Cu concentrations in crab tissue were measured at reference station 3–1 (Fig. 3) and these Cd and Cu concentrations were 2–8 times lower than those reported by Chou et al. (2002) in rock crabs analyzed from stations in the Bay of Fundy (Table 1). The pattern of elevated metal concentrations centred near Muggah Creek observed in sediments (Smith et al., 2009; Walker et al., 2013b), were not observed in crab tissue, and metals in crab tissue and surface sediments were poorly related across all years (r^2 = As, 0.007; Cd, 0.186; Cu, 0.031; Hg, 0.047; Zn, 0.003, $n = 28$). There were no significant differences observed in As, Cd and Cu concentrations between baseline and remediation ($p > 0.134$). Although As concentrations exceeded Canadian Food Inspection Agency (CFIA) guidelines (3.5 μg g^{-1}), As is naturally elevated in rock, soil and sediments across Nova Scotia, which may explain why reference stations exhibited relatively high concentrations (Meunier et al., 2010; CFIA, 2011).

During baseline and year 1 remediation Hg concentrations were ⩽0.01 μg g^{-1}, except for station 1–3 near Muggah Creek during year 1. However, during year 2 and 3 remediation Hg was detected at every station (including reference stations), but were at least an order of magnitude lower than CFIA guidelines (0.5 μg g^{-1}). Where Hg concentrations were below DLs, ½DL was presented (0.005 μg g^{-1}). Mercury concentration during year 1 remediation were not significantly different compared to baseline ($F = 1.00$, $p = 0.337$), but were significantly different during year 2 and 3 (year 2, $F = 54.07$, $p = 0.000$; year 3, $F = 416.53$, $p = 0.000$). Lead concentrations in crabs were below DLs (0.18 μg g^{-1}) for the entire study well below CFIA limits (0.5 μg g^{-1}) (Table 1). Zinc concentrations during year 1 and 3 of remediation were not significantly different compared to baseline (year 1, $F = 0.51$, $p = 0.488$; year 3, $F = 3.10$, $p = 0.109$), but were significantly different during year 2 ($F = 18.51$, $p = 0.001$). Zinc concentrations were lower than those reported by Chou et al. (2002) in their study (24–42 μg g^{-1}) in the Bay of Fundy. In another study, Chou et al. (2000) also reported that Zn accumulation in crabs was typically one-third that found in lobsters indicating an important species difference in the rates of chemical loading.

Metal concentrations in crabs, albeit low, showed little temporal (4 years) and spatial variability, specifically with distance from the STP remediation site, except for a marginal detection of Hg. This lack of spatial difference suggests that the STP site was not the primary source for metals in crab tissue during remediation activities. However, this was not the case with PCBs where minor perturbations were apparent in crabs at near-field stations, although estuarine conditions in

SH may have created a gradient of salinity across stations that may have affected spatial distribution of contaminants (Dean et al., 2009).

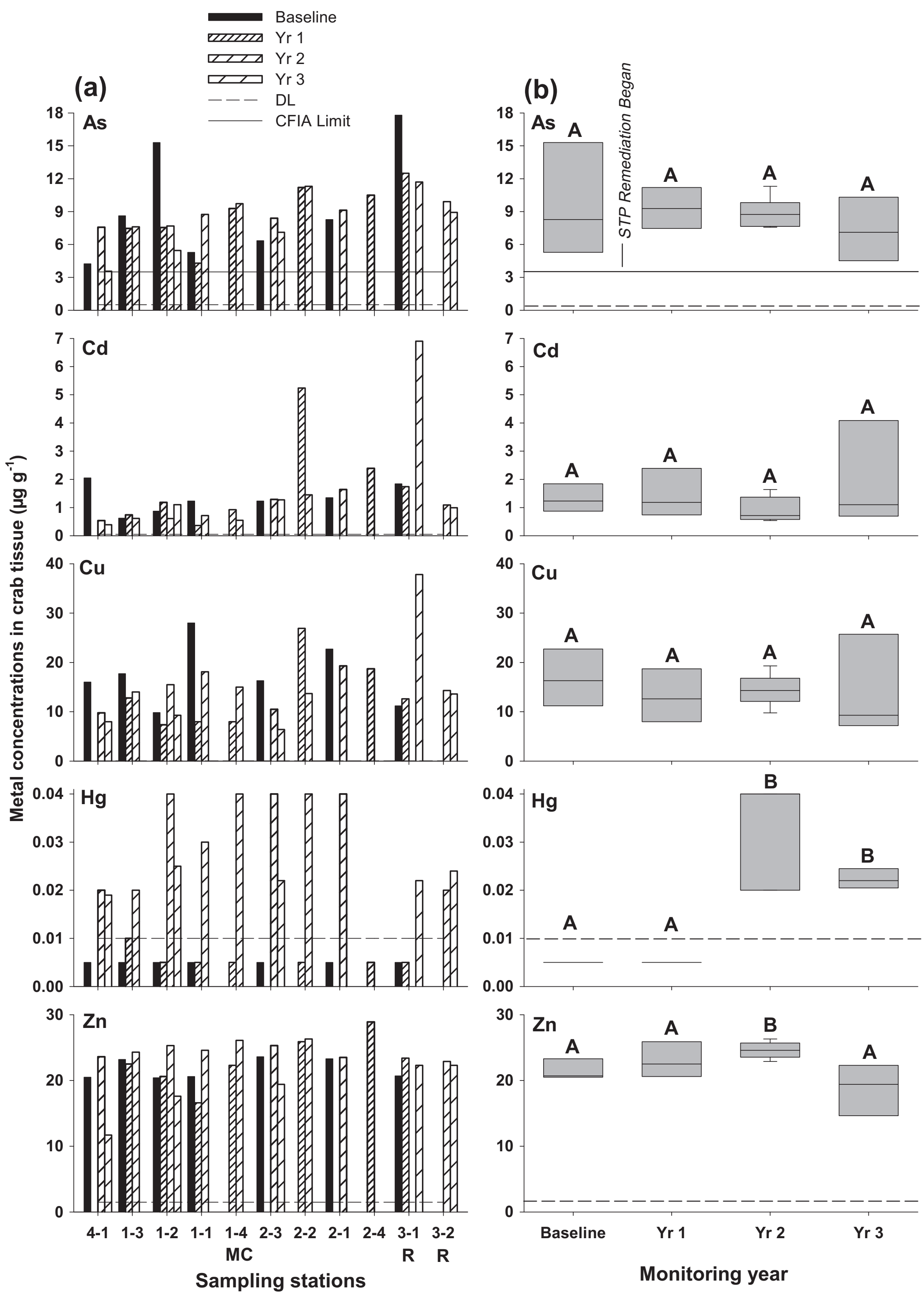

Fig. 3. (a) Concentrations of metals in crabs during baseline and 3 years of remediation. Dashed horizontal lines indicate DLs (As and Cu = 0.5; Cd = 0.005; Hg = 0.01; Zn = 1.5 $\mu g\ g^{-1}$). Solid horizontal lines indicate CFIA guidelines. CFIA limits for Hg (0.5 $\mu g\ g^{-1}$) are not shown due to axis scale. Stations centered on Muggah Creek (MC) and reference stations are indicated (R). (b) Metal concentrations showing temporal variation during baseline and remediation. Significant differences determined using one-way ANOVA followed by Tukey's test; years attributed with same letters were not significant and those with different letters were significantly different ($p < 0.05$).

Lipid contents during baseline ranged between 5.8% and 9.9% (mean 7.6% ± 0.6 SE) and during remediation ranged between 3.7% and 11.0%. The highest lipid content (11%) occurred at stations 4–1 and 1–3 in the inner harbour during year 2 and year 3 remediation. Lipid contents during year 1 and 3 remediation was significantly different compared to baseline ($p < 0.05$), although no significant differences were observed during year 2 remediation ($p > 0.05$). Mean lipid contents for baseline and each subsequent remediation year was 7.6%, 6.7%, 9.0% and 7.0% respectively. These relatively high lipid contents could be an indicator of overall

Table 1
Chemical contaminant concentrations in crustaceans in Nova Scotia (μg g^{-1}).

Species/location	PAH	PCB	As	Cd	Cu	Hg	Pb	Zn	Reference
Rock crabs (*C. irroratus*), SH, NS	<0.05–0.14	0.12–4.5	3.6–15.3	0.5–6.9	9.8–28	<0.01–0.04	<0.18	11.7–28.9	Present study
Canadian guidelines for chemical contaminants and toxins in fish and fish products	–	2	3.5	–	–	0.5	0.5	–	CFIA (2011)
American lobster (*H. americanus*), SH, NS	13–14	–	–	–	–	–	–	–	Ernst et al. (1999)
American lobster (*H. americanus*), SH, NS	0.39–2.24	–	–	–	–	–	–	–	Sirota et al. (1984)
Rock crabs (*C. irroratus*), Bay of Fundy, NS	–	–	–	2.3–187	8.8–520	–	–	19.7–70.4	Chou et al. (2002)
American lobster (*H. americanus*), Bay of Fundy, NS	–	–	–	11.6–22.9	85–896	–	–	28–129	Chou et al. (2000)

For comparison chemical concentrations were derived from hepatopancreas tissues in crabs and lobster.

ecosystem health in Sydney Harbour for this higher trophic level scavenger although concentrations of metals and lipid content in crab tissue were poorly related across all monitoring years (r^2 = 0.022–0.243, n = 28).

Annual sampling of contaminants in crabs in SH has not detected evidence of substantial contaminant releases into the marine environment despite STP remediation activities. Metal concentrations, albeit low, showed little temporal variability, and were not directly related to remediation, with the highest concentrations of As, Cd and Cu measured at far-field stations. Mercury concentrations were at least an order of magnitude lower than CFIA guidelines. Although some moderately elevated PCB concentrations were detected in near-field stations, these were generally lower than those measured during baseline. Natural recovery of SH sediments documented by Smith et al. (2009) and Walker et al. (2013b), may have contributed to a reduction in the contaminant burden in rock crabs. These findings were also corroborated in our results of other sampled ecological media in SH, such as lack of chemical signatures in annual assessments of blue mussel (*Mytilus edulis*) tissue and monthly water quality (Walker et al., 2013c). Whilst current sediment contaminant concentrations are much lower than previously reported (Walker et al., 2013a,b), most of the contaminant inventory remains intact below the sediment surface with the potential risk for future contaminant disturbance or resuspension.

To provide some context related to these findings, PAH, PCB and metal concentrations in crab hepatopancreas were compared to other studies of crab and lobster hepatopancreas, comprising of local (SH) and regional scale (eastern Canada) to better understand the relevance of this study on potential management implications for future Rock Crab Fishing Area and Lobster Fishing Area 27 in Sydney Bight (Schaefer et al., 2004). Overall, current contaminant burdens measured in crabs were much lower than previously reported in other studies of crabs and lobster in industrial harbours in eastern Canada, despite remediation activities (Hilderbrand, 1982; Uthe and Musial, 1986; King et al., 1993; Ernst et al., 1999; King and Chou, 2003). While it is acknowledged that rates of contaminant loading are different between species (Chou et al., 2000), future studies on crabs and lobster in SH may one day lead to the reopening of the commercial lobster fishery in the South Arm of SH.

Acknowledgements

This study was supported by the Sydney Tar Ponds Agency (STPA). We thank Melissa LeRoy, Mark Moriarity, Stuart Sampson and Ellsworth Mailman for help with fieldwork.

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